

the momentum towards making the database available as a service on the cloud. 'Big Data' is changing the traditional ways in which data is stored, accessed and manipulated, with the arrival of the NoSQL movement and domain specific databases, resulting in moving computing closer to data. And if you ask the Internet-savvy the cheeky question, "What is the connection between Cassandra, Voldemort and Dynamo?" (No, this is not related to Harry Potter!), you can expect the majority to give you the correct answer—that they are all specialised NoSQL databases. The fact that databases are no longer the domain of the few specialised expert programmers signifies the shift of the database movement from enterprise-centric workloads to focusing and delivering on the needs of social media networks, collaborative computing, massively multi-player online gaming, etc.

Cloud computing is transforming the way data is stored, retrieved and served. Computing resources like servers, storage, network and applications (including databases) are hosted, and made available as cloud services, for a price. Cloud platforms have evolved to offer many IT needs as online services, without having to invest in expensive data centres and worry about the hassle of managing them. Cloud platforms virtually alleviate the need of having your own expensive data-centre. Today, those managing cloud platforms offer to design, develop, test, deploy and host apps in the cloud environment, with impressively economical cost models. Amazon EC2, Google App Engine, Microsoft Azure, Appistry CloudIQ, AppScale, AT&T Synaptic and RackSpace are just a few of the offerings.

There are various classes of cloud computing services, such as:

- a) **SaaS:** In simple terms, SaaS is a software delivery model wherein business application software is hosted on the Internet/cloud centrally, and can serve multiple customers 'on demand'.
- b) **PaaS:** PaaS provides all systems and environments encompassing the software development life cycle, including development, testing, deploying and the hosting of business applications. Examples include Google App Engine and Microsoft's Azure.
- c) **IaaS:** Infrastructure, in terms of computing resources and operating systems, is also offered as a service.

After all, hosting of business applications on the cloud also entails maintaining and manipulating the data by the applications. Initially, it was left to developers to install, manage and use their choice of database instance on the cloud, with the burden of all the database administration tasks being left to the developer. The advantage of this is that you choose your own database and have full control over how the data is managed. In order to simplify the burden on the users of their cloud offerings, many PaaS vendors today have started offering database services on the cloud. All physical database administration tasks, such as backup, recovery, managing the logs, etc., are managed by the cloud provider. The responsibility for logical administration of the database,

including table tuning and query optimisation, rests on the developer. The choice of databases is also limited typically to MySQL or Oracle RDBMS. Examples include Amazon's Relational Database Service (RDS), Joyent's MySQL (www.joyent.com) and Microsoft's SQL Azure. *Salesforce.com* offers *Database.com*, which is actually built over Oracle's Real Application Clusters. Though *Database.com* itself does not support traditional SQL, the query language it offers is built upon SQL, and is known as 'Salesforce Object Query Language' (SOQL), with limited querying capabilities. This is internally translated to SQL by its system. While these are cloud offerings of database services, their underlying technology is still traditional SQL-based database technology, not specifically re-invented for the cloud.

Of late, there has also been a considerable buzz around offering 'Database as a Service (DaaS)', wherein an organisation's database needs are met by database offerings on the cloud. There are two use-case scenarios:

- (a) A single large organisation that has many individual databases which can be migrated to a private cloud for the organisation, and...
- (b) Outsourcing the data management needs of small and medium organisations to a public cloud provider, who caters to multiple small and medium businesses.

A true DaaS offering should satisfy certain requirements such as:

- (a) Freeing the end developer/user from database administration, tuning and maintenance activities, while offering high performance, availability and fault tolerance, as well as advanced features like snapshot, analytics and time travel.
- (b) Elasticity, or the ability to dynamically adjust to changing workloads. Elasticity is required to meet user SLAs while minimising the cloud provider's infrastructure, power and administration expenses.
- (c) Security and privacy guarantees, and a pay-as-per-usage pricing model. Today, there aren't any true DaaS offerings that satisfy all these requirements. Therefore, these cloud-computing needs will drive the next generation of database evolution.

Big Data explosion and its impact on databases

The term 'Big Data' is used to represent the explosive growth in online data, which has significantly outpaced the increases in CPU processing power, memory and storage capacity over the last few years. For instance, the amount of online data indexed by Google has grown from 5 exabytes in 2002 (1 exabyte is equal to 1 million trillion bytes) to 280 exabytes in 2009, numbers that are conclusively beyond the processing capabilities of any single relational database.

This explosion in data is not just limited to the Web, but has also occurred at the enterprise. Where data was earlier being generated from simple in-house data entry feeds,